PROBLEM REPRESENTED

a. weakness of Ⓡ UL, LL
b. inability of speech

ONSET : Fall

Fall — a. fall caused the weakness/symptom (✗)
b. (OR)
b. ✓ weakness caused fall.

APPROACH TO DI

· FUNCTIONAL disabi
↓
ANATOMICAL G
↓
PATHOLOGY
↓
~~ETIOLOGY~~ PR

I Go fall INJURY FROM FALL

STATIC INJURY (penetrating)
- ICH, Extraaxial Hx (SDH/EDH)
- local INJURIES : Otorrhea, Rhinorrhea, bleeding (epistaxis). Headache.
- Traumatic dissection → intimal tear ⊖

DYNAMIC INJURY
- whiplash injury

II FUNCTIONAL DISABILITY

a. HANDEDNESS : DOMINANT HEMISPHERE

✓ Illuminoscope (Bedside)
Ⓡ handed - Ⓛ ear, Ⓛ foot Ⓡ " - Ⓛ Hemisphere dominant
Ⓛ Handed - Bihemispheric Representation

✓ CLING : both arm together
(dominant thumb ~~at~~ down)

✓ EDINBOROUGH HANDEDNESS INVENTORY
10 POINT - Bedside (≥8/10 = dominant)
21 point - Research

✓ EEG : Normally :
dig ⓐ adult
dominant hemisphere - β. I
Non dom " - relaxation (dominantly α. Rhythm)

✓ Intracarotid amytol (short acting barbiturate) injected [NO LONGER PERFORMED]

✓ functional MRI

ASE DISCUSSION

APPROACH TO DIAGNOSIS

FUNCTIONAL disability
↓
ANATOMICAL CORRELATE
↓
PATHOLOGY
↓
~~ETIOLOGY~~ PROGRESSION : Acute, subacute, ~~progressing~~, relapsing remitting, progression/regression

6. symptom: inability to grip the object (glass of H2O)

jury

✓ grip
- POWER : Holding glass. [LONG FLEXORS]
- PRECISION : Holding pen etc. = FINE MUSCLES [lumbricals, I.ossei]

✓ OTHER POSSIBILITY: SENSORY LOSS → not feeling object → drops the object

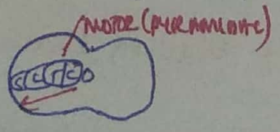
↓
CONCLUSION

MOTOR WEAKNESS :

? LEVEL
- UMN - SPINAL CORD , SUBCORTICAL, CORTICAL
- LMN : muscle, nerve, root, ant. cord

ent
tan
re .
n

UMN - d/t associated speech, LL weakness (MOST PROBABLY)
LMN - could be present
('central cord synd - sulcocommissural A involvement - UL (C > T > L > S))



UMN : CORTEX : seizure, monoparesis, language, cortical sensory
· SUBCORTICAL :

① CONCLUSION : ~~FOR~~ PURELY MOTOR
- LOSS OF POWER GRIP
- UMN TYPE
↓
CORTEX/SUBCORTICAL

✓ able to move limb → fall while trying to walk
↓
BUCKLING → POWER = ≤ 3

· PAINFUL : local cause = cruciate, meniscus injury

PAINLESS : NEURO · PARALYTIC

Ⓝ STANDING : KINETIC ? MELODY ·

✓ gripping chappal difficult : PRECISION GRIP ⊖

② MOTOR → UMN → PRECISION GRIP LOSS
 ↳ (d CORTEX / SUBCORTEX

d. SPEECH vs LANGUAGE

Language : ability of individual to convert thoughts to comprehensible modes of communication
[READ, WRITE, SPEAK, LISTEN,]

SPEECH : CONVERT LANGUAGE TO AUDIBLE SOUNDS

PATIENT : No words spoken,
comprehension Ⓝ, Read Ⓝ write
(illegible words produced)

① ACUTE ANARTHRIA → CAPSULAR GENU SYND: DYSARTHRIA + Clumsy Hand

② MUTISM : SUBCORTICAL LANGUAGE AREAS
[MARIES QUADRILATERAL SPACE]
↓
MOTOR SCHEMA

CHRONIC ANARTHRIA :
✓ MND (MOTOR NEURON disease)
✓ atrophy tongue

③ ANGULAR Branch of MCA POSSIBLE LESIONS
(i) GERSTMANN SYNDROME
(ii) ANARTHRIA
(iii) CONDUCTION APHASIA

CONCLUSION : ⓐ ACUTE ANARTHRIA ⓑ MUTISM ⓒ ANGULAR BR. OF MCA

a. ~~UPPER LIMB MOTOR~~ →

FINAL : ~~MOTOR~~ · MOTOR WEAKNESS [UL ≈ LL] - UMN
↓
SUBCORTICAL
· MUTISM ·
· VASCULAR EVENT - ISCHEMIC (THROMBOTIC)

① HEMORRHAGIC:

- Headache (wake up pt from sleep]   AWARE
- VOMITING
- SEIZURE
- YOUNG →

② THROMBOTIC   EMBOLIC

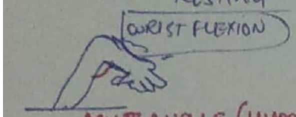
THROMBOTIC	EMBOLIC
✓. Elderly	· young
· gradual progression (Not beyond 72hrs]	· source - ?
	· MAX ONSET
· Headache - may occur early - dull headache	· EXERTION (awake)
	· SEIZURE @ ONSET?
· sleep	- SMALL Hx.
· SEIZURE : scar formation	

emboli
↓
B.V wall ischemia
(↓ relaxation
⇓
stiff vessel)
↓
Blood rush in ← removal by circulation
- shearing force on wall
↓
$\boxed{\text{small Hgh}}$

EXAMINATION

DOMINANT: language, apraxia
NON-": dressing apraxia

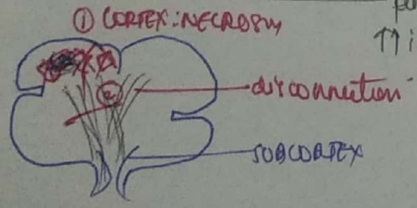
- SPEECH:
  - Broca: 2climbs affected ⊕
- SUBTLE FACIAL PALSY
  - mild asymmetry of palpebral fissure
  - BLINK rate asymmetry [Ⓝ 15-20/min]
  - POWER OF O. OCULI, O. ORIS, BUCCINATOR.
- HYPOTONIA
  - MOST SUBTLE SPASTICITY
    - UL: PRONATORS
    - LL: ADDUCTORS.
  - Shake limb
    - >3 movements/oscillation
- ELBOW RESTING



ACUTE ANGLE (HYPOTONIA) [∠]
OBTUSE ANGLE (∟) ↑TONE.

- POWER:
  - ✓ distal > proximal (UMN)
  - ✓ UL > LL (MCA)

Why pyramidal pattern is seen during recovery? (UL>LL



→ antigravity > ↑↑ involved)

BASIC = FUNCTION = SURVIVAL
- EATING
- RUNNING (LOCATION)

CVA: loss of cortical volition (loss of connection/
↓
PRIMITIVE FUNCTION ~~loss~~ ⊕
↓
(UL FLEXION: EATING
~~LOU~~
LL EXTENSION: LOCOMOTION

REFLEXES

① absent Abdominal reflex

	UMN	LMN (radiculopathy)
✓ superficial Ab reflex	LOST	LOST
✓ DEEP ABD REFLEX	Exaggerated	LOST

② SUBTLE PLANTAR ↑E

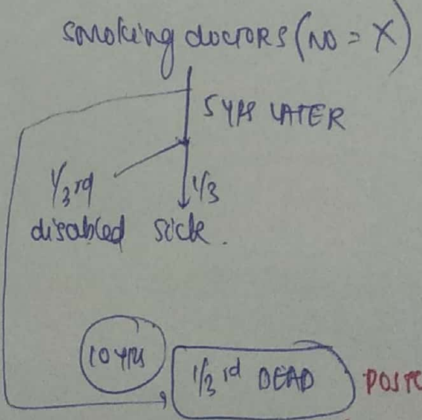
. PURELY MOTOR, c̄ ANARTHRIA / MUTISM

1. MIDDLE CEREBRAL BRANCHES - ANGULAR BRANCHES
2. DYSARTHRIA c̄ CLUMSY HAND SYNDROME [LACUNAR SYNDROME]

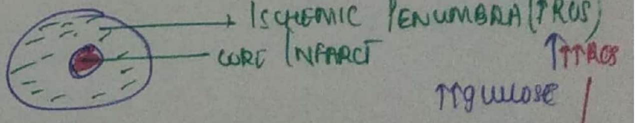
RARE CASES: I/L LESION + I/L HEMIPLEGIA / HEMISENSORY LOSS
- SUPPLEMENTARY MOTOR AREA ⊕ (AREA 6)
- " SENSORY AREA

PECULIARITY
B/L Representation
Hence ⓐ I/L / C/L / Bilateral loss

SMOKING STUDY



POSTULATED (Extrapolated to have deaths like in World War



↑↑ glucose

Hence strict glucose control

EXPANSION OF PENUMBRA